

Ariun-Erdene Tumurchuluun, CV

CONTACT

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FUNDING AND SCHOLARSHIPS

2023 – 2025 **Erasmus Mundus Scholarship**

Joint master's scholarship in Language and Communication Technologies (LCT); stipend, tuition, transportation, insurance, and related allowances.

EDUCATION

Oct 2024 – Sep 2025 **M.Sc., Language Science and Technology**, *Universität des Saarlandes*, Saarbrücken, Germany

Thesis: *Causal Investigation of Tense Encoding in a Multilingual Transformer*.

Final grade: 1.7 / 5.0 (German scale; 1.0 = best).

Selected coursework: Mechanistic Interpretability, Multimodal Language Understanding, Statistical NLP, Speech Science, Speech Recognition and Generation, Software Project (mechanistic approaches in NLP).

Oct 2023 – Sep 2025 **M.Sc., Language Technologies and Computational Linguistics**, *Charles University*, Prague, Czech Republic

Thesis: *Causal Investigation of Tense Encoding in a Multilingual Transformer*.

GPA: 1.91 / 4.0 (Czech scale; 1.0 = best), Summa Cum Laude.

Selected coursework: Deep Learning, Unsupervised Machine Learning, Statistical Methods in NLP I & II, Recent Advances in Mechanistic Interpretability, Multimodal Language Understanding.

Sep 2018 – May 2022 **B.Eng., Software Engineering**, *National University of Mongolia*, Ulaanbaatar, Mongolia

GPA: 3.8 / 4.0 (top 1%), Summa Cum Laude.

Selected coursework: Algorithms, Data Structures, Mathematical Logic and Discrete Mathematics, Machine Learning, Pattern Recognition, Databases.

SELECTED RESEARCH PROJECTS

Nov 2025 – Present **Tracing Stereotypes from Representation to Output in Multilingual LLMs**

Investigated whether stereotype bias representations in state-of-the-art LLMs are shared across languages or language-specific across model families. Designed analyses spanning gender, age, sexual orientation, physical appearance, and related attributes using the MBBQ and BiasShades benchmarks. Accepted as a main conference paper at EMNLP 2026.

Jul 2026 – Aug 2026 **Emergent Misalignment Across Model Scale and Data Domain**

BlueDot Technical AI Safety Project. Studied emergent misalignment in two model sizes, fine-tuned on narrowly misaligned prose-advice and code datasets. Bad prose advice generalized to broadly misaligned behaviour, and the components responsible were localised. In contrast, bad code did not transfer directly to misaligned prose, leaving open whether the models were simply too small for the effect to emerge. [Link to blog](#).

Jun 2025 – Oct 2025 **TenseLoC: Tense Localization and Control in a Multilingual LLM**

Studied internal representations of grammatical tense (morphology, syntax, semantics) in Llama 3.1 across eight typologically diverse languages, using probing, causal tracing, sparse autoencoders, and activation steering grounded in identified tense features. Best Paper Award at MRL @ EMNLP 2025.

Apr 2025 **Word-Order Encoding in Multilingual BERT**

Implemented diagnostic classifiers to quantify mBERT sensitivity to word-order perturbations across six typologically diverse languages.

INDUSTRY EXPERIENCE

- Nov 2025 – May 2026 **AI Engineer**, *Egune AI LLC*, Ulaanbaatar, Mongolia
Built NLP systems in Mongolian for spelling and grammar correction, POS tagging, word sense and homonym disambiguation, translation evaluation, RAG-based document retrieval, and text rewriting. Curated and generated training data; fine-tuned models with full fine-tuning, LoRA, and QLoRA using Axolotl; qualitative and quantitative evaluation; inference with Hugging Face Transformers and vLLM; API development and demo documentation; data collection with Playwright.
- Jun 2021 – Jul 2023 **Software Engineer**, *AI Lab LLC*, Ulaanbaatar, Mongolia
Full-stack engineer, delivered and maintained three production web applications (investment and crowdfunding); implemented microservice features, search improvements, and OAuth SSO (Firebase Authentication, Google Cloud Functions); migrated identity infrastructure from Keycloak to Firebase Auth; maintained a Django crowdfunding platform. Stack: Python, PHP, PostgreSQL, AWS, Firebase, Redis, HTML/CSS/JavaScript, automated unit testing.

PUBLICATIONS AND MANUSCRIPTS

Peer-reviewed

1. **Ariun-Erdene Tumurchuluun***, Yusser Al Ghussin*, Pinzhen Chen, Josef Van Genabith, and Koel Dutta Chowdhury. 2026. *Tracing Stereotypes from Representation to Output in Multilingual LLMs*. To appear in *Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, Main Conference (*equal contribution).
2. **Ariun-Erdene Tumurchuluun**, Yusser Al Ghussin, David Mareček, Josef Van Genabith, and Koel Dutta Chowdhury. 2025. *TenseLoC: Tense Localization and Control in a Multilingual LLM*. In *Proceedings of the 5th Workshop on Multilingual Representation Learning (MRL)*, pages 243–264, Suzhou, China. Association for Computational Linguistics. **Best Paper Award**.

Preprints

1. Pinzhen Chen, Koel Dutta Chowdhury, **Ariun-Erdene Tumurchuluun** and collaborators. 2026. *Cultivar: A Contrastive and Locale-Oriented Translation Benchmark for Investigating Contamination and Localisation Robustness*. arXiv:2608.09766. Contributed the Khalkha Mongolian portion. Collaboration with Queen's University Belfast and Saarland University.

AWARDS AND HONOURS

- Best Paper Award, 5th Workshop on Multilingual Representation Learning (MRL) @ EMNLP, 2025
- Most Advanced Graduate Award, National University of Mongolia, 2022

CERTIFICATIONS

- Jul 2026 – Aug 2026 **Technical AI Safety Project**, *BlueDot Impact*
Independent research sprint on AI Safety: Studied emergent misalignment in LLMs by fine-tuned 0.5B and 14B models on narrowly misaligned prose-advice and code data and traced where the resulting misalignment is represented. Certificate ID: recLEDmMnJbRZlz0o.
- Jun 2026 **Technical AI Safety**, *BlueDot Impact*
30-hour program on technical AI safety, covering current safety techniques, threat modelling via kill-chain analysis, and interpretability (“Understanding AI”). Certificate ID: recJARgW9cpMYJ0o7.

TECHNICAL SKILLS

- ML & NLP** Python, PyTorch, Tensorflow, Hugging Face Transformers, vLLM; fine-tuning with full, LoRA, and QLoRA (Axolotl, ms-swift); evaluation and error analysis; data curation; RAG pipelines; mechanistic interpretability tools (probing, causal tracing, attribution patching, sparse autoencoders, activation steering).
- Engineering** Git, Linux, GPU clusters, visualization, unit and integration testing, REST APIs, PostgreSQL, Redis, Django, Vue.js, cloud backends (AWS, Firebase, Google Cloud Functions), Playwright.
- Languages** English (professional); Mongolian (native); German, French, Russian, Japanese (beginner).